

$$P\left(-z_{\frac{\alpha}{2}} < \frac{\bar{X}_n - \mu}{S/\sqrt{n}} < z_{\frac{\alpha}{2}}\right) \approx 1-\alpha \quad \text{if } n \text{ is large enough.}$$

$\bar{X}_n \pm z_{\frac{\alpha}{2}} \cdot \frac{S}{\sqrt{n}}$ is an approximate $1-\alpha$ level CI for μ . This holds also in the non-normal case.

What happens when n is not large enough?

$$\text{If } X_1, \dots, X_n \sim N(\mu, \sigma)$$

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \sim t_{n-1} \quad \left(t\text{-distribution with degree of freedom } n-1 \right)$$

$$P\left(-t_{\frac{\alpha}{2}}(n-1) < \frac{\bar{X}_n - \mu}{S/\sqrt{n}} < t_{\frac{\alpha}{2}}(n-1)\right) = 1-\alpha$$

$\bar{X}_n \pm t_{\frac{\alpha}{2}}(n-1) \cdot \frac{S}{\sqrt{n}}$ is an (exact) CI for μ at level $1-\alpha$.